Dr. Alexander von Humboldt

Distinguished Research Scientist & Distributed Systems Architect

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Kurzprofil

Pioneering researcher and systems engineer with over 15 years of experience leading high-performance computing initiatives across European research networks.

Berufserfahrung

Research Lab Europe 1, Principal Systems Architect 1 — Berlin / Munich — Jan 2011 – Dez 2012

- Architected high-throughput parallel compute cluster for astronomical data analysis across tier-1 data centers.
- Optimized memory bus saturation reducing compute cycle bottlenecks by 31%.
- Implemented lock-free asynchronous queue topologies handling 40M events/sec.
- Co-authored patent on distributed zero-copy network buffer management.

Research Lab Europe 2, Principal Systems Architect 2 — Berlin / Munich — Jan 2012 – Dez 2013

- Architected high-throughput parallel compute cluster for astronomical data analysis across tier-1 data centers.
- Optimized memory bus saturation reducing compute cycle bottlenecks by 32%.
- Implemented lock-free asynchronous queue topologies handling 40M events/sec.
- Co-authored patent on distributed zero-copy network buffer management.

Research Lab Europe 3, Principal Systems Architect 3 — Berlin / Munich — Jan 2013 – Dez 2014

- Architected high-throughput parallel compute cluster for astronomical data analysis across tier-1 data centers.
- Optimized memory bus saturation reducing compute cycle bottlenecks by 33%.
- Implemented lock-free asynchronous queue topologies handling 40M events/sec.
- Co-authored patent on distributed zero-copy network buffer management.

Research Lab Europe 4, Principal Systems Architect 4 — Berlin / Munich — Jan 2014 – Dez 2015

- Architected high-throughput parallel compute cluster for astronomical data analysis across tier-1 data centers.
- Optimized memory bus saturation reducing compute cycle bottlenecks by 34%.
- Implemented lock-free asynchronous queue topologies handling 40M events/sec.
- Co-authored patent on distributed zero-copy network buffer management.

Research Lab Europe 5, Principal Systems Architect 5 — Berlin / Munich — Jan 2015 – Dez 2016

- Architected high-throughput parallel compute cluster for astronomical data analysis across tier-1 data centers.
- Optimized memory bus saturation reducing compute cycle bottlenecks by 35%.
- Implemented lock-free asynchronous queue topologies handling 40M events/sec.
- Co-authored patent on distributed zero-copy network buffer management.

Ausbildung

Doctor of Science — **Technical University of Berlin 1**, Computer Science & Parallel Architectures — Berlin, Germany — Okt 2001 – Jul 2004

- GPA / Honors: Summa Cum Laude
- Dissertation on deterministic latency guarantees in multi-tenant cloud fabrics.

Master of Science — **Technical University of Berlin 2**, Computer Science & Parallel Architectures — Berlin, Germany — Okt 2002 – Jul 2005

- GPA / Honors: Summa Cum Laude
- Dissertation on deterministic latency guarantees in multi-tenant cloud fabrics.

Kenntnisse & Fähigkeiten

Core Architecture: Distributed Systems, C++20, Rust

Infrastructure: Kubernetes

High Performance Computing: MPI / OpenMP, RDMA InfiniBand

Projekte

HPC-Orchestrator-1 Jan 2021 – Jan 2023

Rust, Tokio, gRPC, eBPF

- Kernel-bypass networking scheduler for heterogeneous accelerator nodes.

HPC-Orchestrator-2 Jan 2021 – Jan 2023

Rust, Tokio, gRPC, eBPF

- Kernel-bypass networking scheduler for heterogeneous accelerator nodes.

HPC-Orchestrator-3 Jan 2021 – Jan 2023

Rust, Tokio, gRPC, eBPF

- Kernel-bypass networking scheduler for heterogeneous accelerator nodes.

Sprachen

Sprachen: Deutsch (Muttersprache), English (Fließend), Français (Gute Kenntnisse)